

Networking Assignment Week 3

Capture and analyze ARP packets using Wireshark. Inspect the ARP request and reply frames, and discuss the role of the sender's IP and MAC address in these packets.

No.	Time	Source	Destination	Protocol	Length	Info
763	56.590482396	20:58:43:e6:8e:bc	DZS_c5:bc:cf	ARP	42	Who has 192.168.1.1? Tel
764	56.591990048	DZS_c5:bc:cf	20:58:43:e6:8e:bc	ARP	42	192.168.1.1 is at 30:4f:
1203	96.526568270	20:58:43:e6:8e:bc	DZS_c5:bc:cf	ARP	42	Who has 192.168.1.1? Tel
1205	96.532559614	DZS_c5:bc:cf	20:58:43:e6:8e:bc	ARP	42	192.168.1.1 is at 30:4f:

▶	Frame 763: 42 bytes on wire (336 bits), 42 bytes captured	0000	30 4f 75 c5 bc cf 20 58 43 e6 8e bc
▶	Ethernet II, Src: 20:58:43:e6:8e:bc (20:58:43:e6:8e:bc),	0010	08 00 06 04 00 01 20 58 43 e6 8e bc
▼	Address Resolution Protocol (request)	0020	00 00 00 00 00 00 c0 a8 01 01
	Hardware type: Ethernet (1)		
	Protocol type: IPv4 (0x0800)		
	Hardware size: 6		
	Protocol size: 4		
	Opcode: request (1)		
	Sender MAC address: 20:58:43:e6:8e:bc (20:58:43:e6:8e:bc)		
	Sender IP address: 192.168.1.5		
	Target MAC address: 00:00:00:00:00:00 (00:00:00:00:00:00)		
	Target IP address: 192.168.1.1		

No.	Time	Source	Destination	Protocol	Length	Info
763	56.590482396	20:58:43:e6:8e:bc	DZS_c5:bc:cf	ARP	42	Who has 192.168.1.1? T
764	56.591990048	DZS_c5:bc:cf	20:58:43:e6:8e:bc	ARP	42	192.168.1.1 is at 30:4
1203	96.526568270	20:58:43:e6:8e:bc	DZS_c5:bc:cf	ARP	42	Who has 192.168.1.1? T
1205	96.532559614	DZS_c5:bc:cf	20:58:43:e6:8e:bc	ARP	42	192.168.1.1 is at 30:4

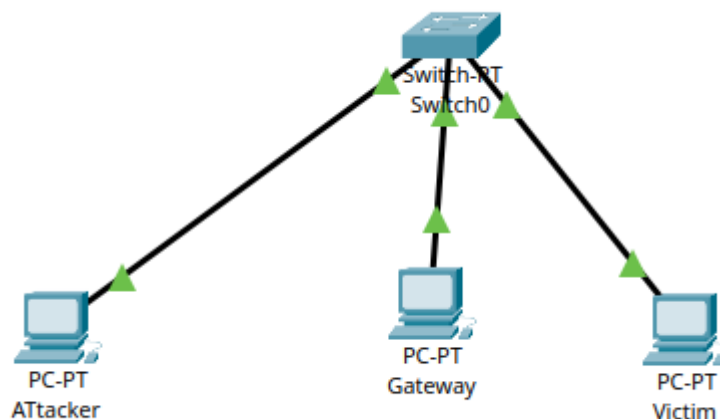
▶	Frame 764: 42 bytes on wire (336 bits), 42 bytes captured	0000	20 58 43 e6 8e bc 30 4f 75 c5 bc
▶	Ethernet II, Src: DZS_c5:bc:cf (30:4f:75:c5:bc:cf), Dst:	0010	08 00 06 04 00 02 30 4f 75 c5 bc
▼	Address Resolution Protocol (reply)	0020	20 58 43 e6 8e bc c0 a8 01 05
	Hardware type: Ethernet (1)		
	Protocol type: IPv4 (0x0800)		
	Hardware size: 6		
	Protocol size: 4		
	Opcode: reply (2)		
	Sender MAC address: DZS_c5:bc:cf (30:4f:75:c5:bc:cf)		
	Sender IP address: 192.168.1.1		
	Target MAC address: 20:58:43:e6:8e:bc (20:58:43:e6:8e:bc)		
	Target IP address: 192.168.1.5		

- ARP packets were captured using Wireshark.

- An ARP request packet was observed where the sender IP 192.168.1.5 asked for the MAC address of 192.168.1.1.
- The request was broadcast to all devices in the network.
- An ARP reply was received from 192.168.1.1 with its MAC address.
- The reply was unicast and sent directly to the requester.
- The sender MAC and IP identify the source device, and the target IP is the address being resolved.
- This demonstrates how ARP is used to map IP addresses to MAC addresses for communication.

Using Packet Tracer, simulate an ARP spoofing attack. Analyze the behavior of devices on the network when they receive a malicious ARP response.

- A network with attacker, victim, and gateway was created using a switch.
- Initially, the victim system communicated with the gateway and stored its MAC address in the ARP table.
- The attacker system then impersonated the gateway by using the same MAC address.
- A fake ARP response was generated and sent to the victim.
- After simulation, the victim's ARP table was observed.
- It was found that the attacker IP address was mapped to the gateway's MAC address.
- This shows that the victim was misled into associating the attacker with the gateway.
- This demonstrates ARP poisoning, where incorrect MAC-IP mapping redirects traffic through the attacker.



The image shows a Cisco Packet Tracer interface. The top window is titled 'Victim' and has tabs for Physical, Config, Desktop, Programming, and Attributes. The 'Desktop' tab is active, showing a 'Command Prompt' window. The Command Prompt displays the output of a ping command to 192.168.1.1, showing successful results with 0% loss. Below the ping output, the 'arp -a' command is executed, showing the ARP table with entries for 192.168.1.1 and 192.168.1.20, both with physical address 0007.ec5a.ad46 and type 'dynamic'.

Below the Command Prompt window is the 'Simulation Panel'. It contains an 'Event List' table with the following data:

Vis.	Time(sec)	Last Device
	1.000	--
	1.001	Attacker
	1.002	Switch0
	1.003	Victim
	1.004	Switch0

Below the Event List table is a terminal window showing the output of the 'arp -a' command, which is identical to the one shown in the Command Prompt window above.

Manually configure static IPs on the client devices(like PC or your mobile phone) and verify connectivity using ping.

```
manish@boss:~$ sudo ip addr add 192.168.1.100/24 dev wlo1
manish@boss:~$ ip addr show wlo1
2: wlo1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 20:58:43:e6:8e:bc brd ff:ff:ff:ff:ff:ff
    altname wlp2s0
    inet 192.168.1.5/24 brd 192.168.1.255 scope global dynamic noprefixroute wlo1
        valid_lft 84926sec preferred_lft 84926sec
    inet 192.168.1.100/24 scope global secondary wlo1
        valid_lft forever preferred_lft forever
    inet6 2401:4900:1ce0:1424:ee49:a47b:1e6a:a9e7/64 scope global dynamic noprefixroute
        valid_lft 86367sec preferred_lft 86367sec
    inet6 fe80::3745:56f:2c7f:63fd/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

- A static IP address was manually configured on a Linux system using the ip command.
- The IP address was assigned using:
sudo ip addr add 192.168.1.100/24 dev wlo1
- The assigned IP address was verified using:
ip addr show wlo1
- The output shows both the original IP (192.168.1.5) and the newly assigned static IP (192.168.1.100).

Use Wireshark to capture DHCP Discover, Offer, Request, and Acknowledge messages and explain the process.

- DHCP packets were captured using Wireshark.
- The DHCP process follows four steps known as DORA: Discover, Offer, Request, and Acknowledge.
- In this capture, DHCP Request and Acknowledge packets were observed.
- The client sent a DHCP Request message to renew or confirm its IP address.
- The DHCP server responded with an Acknowledge message, confirming the assigned IP address.
- The ACK packet contains the client IP address and network configuration details.
- This indicates DHCP renewal, where the client already had an IP address and requested confirmation.
- This demonstrates how DHCP dynamically manages IP address allocation in a network.

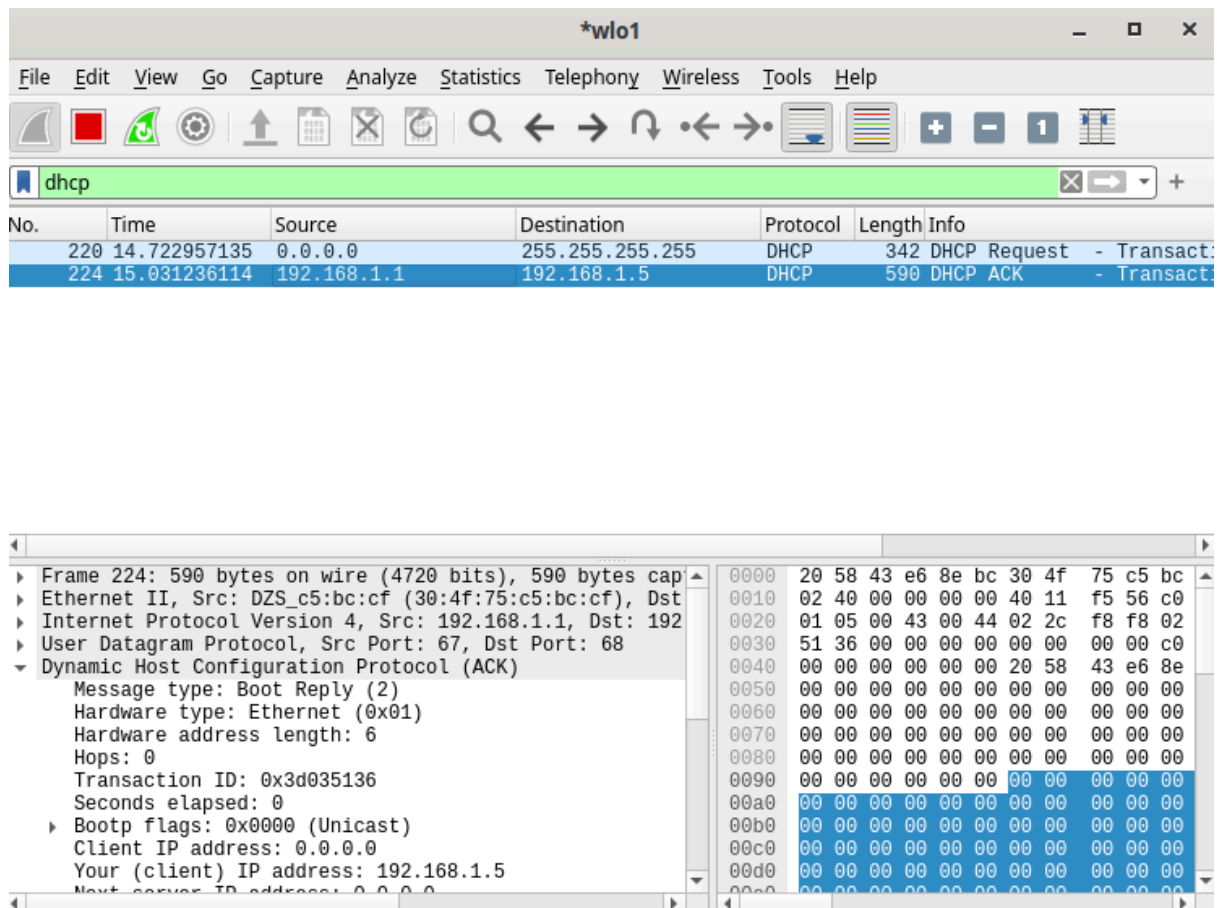
The screenshot displays the Wireshark interface with the network interface **wlo1** selected. The packet list shows two DHCP packets:

No.	Time	Source	Destination	Protocol	Length	Info
220	14.722957135	0.0.0.0	255.255.255.255	DHCP	342	DHCP Request - Transaction ID 0x3d035136
224	15.031236114	192.168.1.1	192.168.1.5	DHCP	590	DHCP ACK - Transaction ID 0x3d035136

The packet details for Frame 220 (DHCP Request) are expanded, showing the following information:

- Dynamic Host Configuration Protocol (Request)
 - Message type: Boot Request (1)
 - Hardware type: Ethernet (0x01)
 - Hardware address length: 6
 - Hops: 0
 - Transaction ID: 0x3d035136
 - Seconds elapsed: 0
 - Bootp flags: 0x0000 (Unicast)
 - Client IP address: 0.0.0.0
 - Your (client) IP address: 0.0.0.0

The packet bytes pane shows the raw data of the DHCP request, with the first 12 bytes (0000 ff ff ff ff ff ff 20 58 43 e6 8e) highlighted in blue.



Given an IP address range of 192.168.1.0/24, divide the network into 4 subnets. Task: Manually calculate the new subnet mask and the range of valid IP addresses for each subnet. Assign IP addresses from these subnets to devices in Cisco Packet Tracer and verify connectivity using ping between them.

The original /24 subnet (255.255.255.0) has 256 total addresses (0-255).

We need 4 subnets, so we borrow 2 bits from the host portion.

New Subnet Mask: /26 (255.255.255.192) Each subnet has 64 addresses ($2^6 = 64$). Usable IPs per subnet: 62 (since 2 are reserved for network ID and broadcast).

Subnet 1: 192.168.1.0/26 → Valid IPs: 192.168.1.1 to 192.168.1.62 (Broadcast: 192.168.1.63)

Subnet 2: 192.168.1.64/26 → Valid IPs: 192.168.1.65 to 192.168.1.126 (Broadcast: 192.168.1.127)

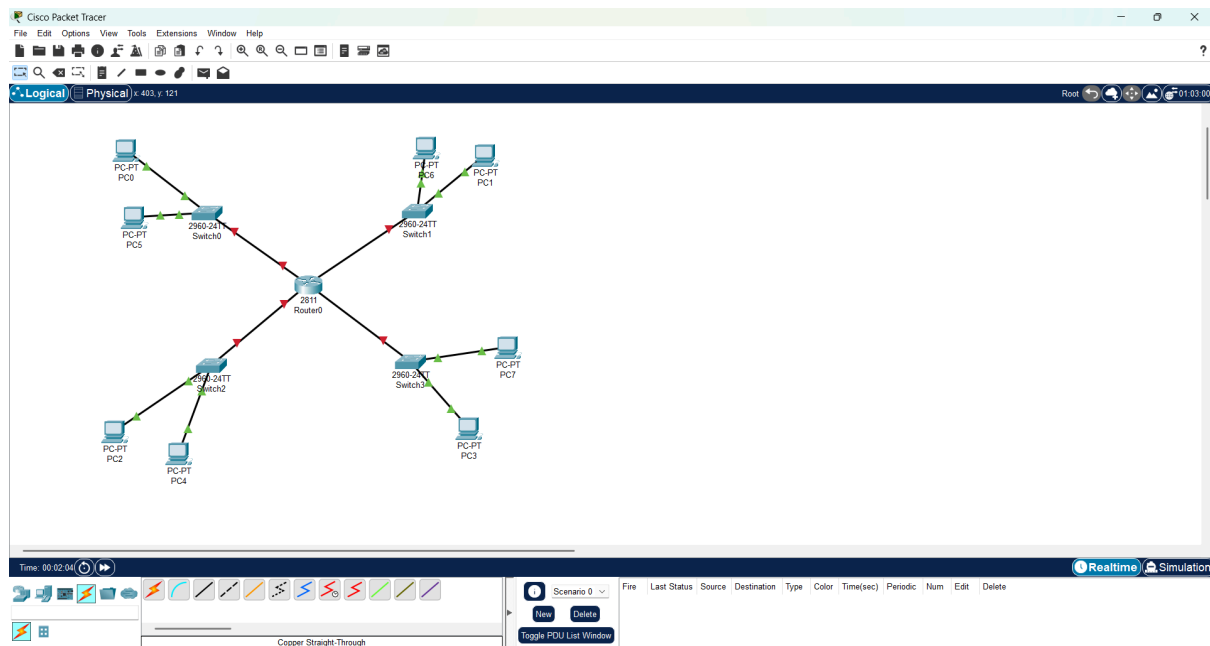
Subnet 3: 192.168.1.128/26 → Valid IPs: 192.168.1.129 to 192.168.1.190 (Broadcast: 192.168.1.191)

Subnet 4: 192.168.1.192/26 → Valid IPs: 192.168.1.193 to 192.168.1.254 (Broadcast: 192.168.1.255)

Implementation in Cisco Packet Tracer:

We need to create four separate LAN segments using switches.

Then we need to Assign static IPs to devices in each subnet.



You are given three IP addresses: 10.1.1.1, 172.16.5.10, and 192.168.1.5. Task: Identify the class of each IP address (Class A, B, or C). What is the default subnet mask for each class? Provide the range of IP addresses for each class.

IP Address Analysis:

10.1.1.1:

Class: A

Default Subnet Mask: 255.0.0.0 (/8)

Range (Class A): 1.0.0.0 – 126.255.255.255

172.16.5.10:

Class: B

Default Subnet Mask: 255.255.0.0 (/16)

Range (Class B): 128.0.0.0 – 191.255.255.255

192.168.1.5:

Class: C

Default Subnet Mask: 255.255.255.0 (/24)

Range (Class C): 192.0.0.0 – 223.255.255.255